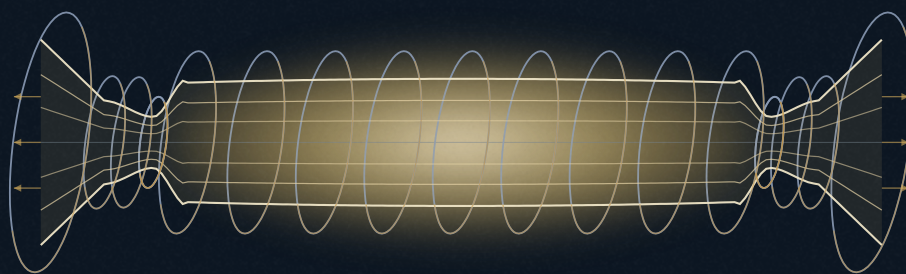




KRONOS FUSION ENERGY



PAPER 2.34 OF THE 2026 KRONOS SET

# *The HVDC-Native Power Interface*

## *MetroVolt to the Data-Center Bus*

A direct-energy-conversion machine is intrinsically an HVDC source — a shorter, different electrical interface.

~54% DC-native at 50-250 kV · 800 VDC in 2-3 stages · ms load-following

Explore it live — [3-D model](#) · [physics validator](#) · [film series](#)



THE OFFICIAL RECORD

# What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

|                 |                                                                                                                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TITLE           | The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study                                                                                                                                                                       |
| AUTHOR          | Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy                                                                                                                                                                                                    |
| EDITION         | 2026 Combined Editorial · first issue                                                                                                                                                                                                                              |
| COMPANION WORKS | The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop. |
| NAMING          | Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.                                                                                               |
| STATUS          | Conceptual design & simulation study. Nothing herein is a construction commitment.                                                                                                                                                                                 |



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

|              | HYPERION                             | AEGIS                        | METROVOLT                       |
|--------------|--------------------------------------|------------------------------|---------------------------------|
| Role         | Strategic-isotope foundry            | Defense installation power   | Campus power                    |
| Machine      | Spherical tokamak                    | Tandem mirror                | Tandem mirror                   |
| Fuel         | Deuterium–tritium                    | Deuterium–helium-3           | Deuterium–helium-3              |
| Delivers     | Tritium · <sup>3</sup> He · 14 MeV n | Resilient installation power | Firm campus power               |
| Physics bar  | Fusion gain only                     | Closure (gates named)        | Closure + lunar <sup>3</sup> He |
| In the fleet | Breeds the fuel                      | Proves the generator         | Commercial destination          |

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

## ABSTRACT

# The HVDC-Native Power Interface — MetroVolt to the Data-Center Bus

A fusion generator built around direct energy conversion is intrinsically a high-voltage DC source. About 54% of gross output leaves the converter as DC at 50–250 kV, reaching an 800 V rack bus in 2–3 power-electronic stages (vs 4–5 for an AC plant). Load-following belongs on the ms-class electronic path: a 4.14 gross/net amplifier makes plasma throttling expensive. Economics/siting held internal.

Companion deposit (code & data, reproducible under CC BY 4.0): DOI [10.5281/zenodo.22132211](https://doi.org/10.5281/zenodo.22132211) · Zenodo community *kronos\_fusion\_energy*.

Explore it live: [interactive 3-D model](#) · [run the physics validator](#) · [companion film series](#) (the deposited solvers, in the browser).

**Keywords:** [direct energy conversion](#) [HVDC](#) [power electronics](#) [load following](#)

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NOMENCLATURE

# Symbols, units, and the names of things

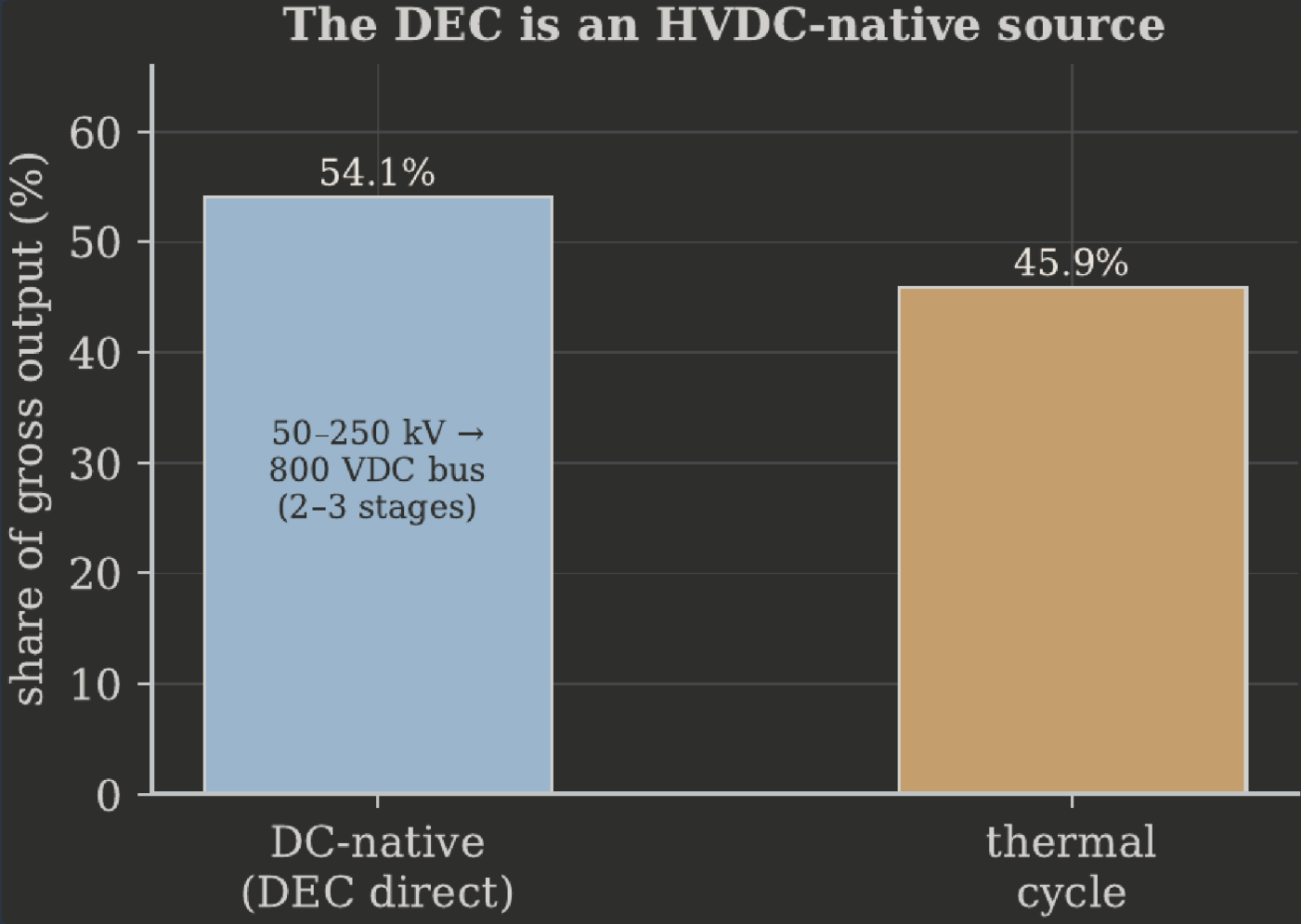
|                  |                                                                        |
|------------------|------------------------------------------------------------------------|
| $Q$              | plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$                  |
| $Q_E$            | engineering gain, net electric out / recirculating in                  |
| $H_{98}$         | confinement enhancement over the IPB98(y,2) scaling                    |
| $Z_{\text{eff}}$ | effective ion charge                                                   |
| $c_{ee}$         | electron–electron bremsstrahlung leading coefficient, 2.120022 (exact) |
| $\tau_E$         | energy confinement time                                                |
| $I_p$            | plasma current                                                         |
| $R_0, a$         | major radius, minor radius                                             |
| $\kappa, \delta$ | elongation, triangularity                                              |
| $\beta_N$        | normalized plasma beta (Troyon-normalized)                             |
| $TBR$            | tritium breeding ratio                                                 |
| $DEC$            | direct energy conversion                                               |
| $CF$             | plant capacity factor (availability)                                   |
| $FOAK/NOAK/BOAK$ | first / n <sup>th</sup> / best of a kind                               |
| <i>Hyperion</i>  | the ST breeder (internal: Mode D2)                                     |
| <i>Aegis</i>     | the generator, defense/installation housing (internal: Mode M)         |
| <i>MetroVolt</i> | the generator, data-center housing (Mode M)                            |

# Not an AC machine in disguise

Most fusion power-plant studies assume an AC turbine hall. A direct-energy-conversion machine does not have one: its converters produce DC at high voltage. When the load is a data center — itself a DC bus — the interface is shorter and different.

### AN HVDC-NATIVE SOURCE

About 54% of gross output is delivered as DC at 50–250 kV, with no synchronous generator and no grid-frequency stage. Reaching an 800 V rack bus takes 2–3 stages, against 4–5 for a conventional AC plant that must synchronize then rectify.



**FIGURE 1.** The direct-energy converter delivers ~54% of gross output as HVDC (gate MV-L1-A3). ARCHITECTURE



## § 2

## Follow the load with electronics, not plasma

---

A direct converter modulates at ms scale. Following the load by throttling the plasma is far more costly: the gross-to-net amplifier is 4.14, so a 10% plasma dip cuts net export ~41%. Load-following is done on the electronic path with dump and storage, holding the plasma at its efficient point. Economics, siting, and service-level content are internal and excluded.

**POWER ELECTRONICS**

A P P A R A T U S

## *Portfolio, People & Record*

*The intellectual-property estate, the people who built it, the limitations  
that gate it, and the sources it rests on.*

1

GRANTED PATENT

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INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

|               |                                                                                   |         |
|---------------|-----------------------------------------------------------------------------------|---------|
| US 12,009,112 | High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550 | GRANTED |
| US 17/878,507 | Core device / method (KRONO-001A) · utility                                       | PENDING |

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

|            |                                                                                                                                                                                                                                |       |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 64/124,209 | Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026                                                                                                                                             | FILED |
| 64/124,220 | Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026                                                                                                                             | FILED |
| 64/115,167 | KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026                                                                                                                                                          | FILED |
| 64/005,440 | Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026                                                                                                                       | FILED |
| 64/105,530 | Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026                                                                                                                                | FILED |
| 64/128,097 | Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop | FILED |

TRADEMARK & ORIGIN

|              |                                                                          |          |
|--------------|--------------------------------------------------------------------------|----------|
| FTK-98801894 | Kronos Fusion Energy — trademark application                             | FILED    |
| KR-2022-★    | Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022 | PRIORITY |

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any  $Q > 1$  milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

## THE TEAM

## Founding partners, board, advisors &amp; auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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Legal counsel is retained; those roles are held on the internal roster and are not listed here.

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## COLOPHON

# How this document was made

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*The Kronos Fleet — Combined Editorial, 2026* is set in **Fraunces** (display), **Gelasio** (text), and **IBM Plex Mono** (data & equations), carried forward from the Kronos editorial design system.

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The figures are rendered at 300 dpi from the deposited generator scripts; the document is composed as a single self-contained file with embedded fonts and imagery, paginated to US Letter, and rendered to PDF through a headless Chromium engine in matched light and dark editions.

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Every headline quantity carries an evidence class; withdrawn and restated values are marked as such. Nothing herein is frozen beyond the founder-approved Tier-A set.

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**Numbers and their classes.** Quantities are reported with evidence classes and case labels; modelled, assumed, and requirement-class values are identified as such and are not to be quoted without their labels.

**Public edition.** This edition carries the physics and design only; all financial, funding, and defense-commercial content has been removed for public distribution. The scientific text corresponds to the arXiv and journal submissions.

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THE 2026 KRONOS SET

# One programme, in five papers

This editorial is one of five that together describe the Kronos fusion programme — a breeder that funds a generator, and the magnet, conversion, and control science that enable both. Each is published open-access with a reproducible companion deposit.

|           |                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------|
| PAPER I   | Hyperion — The Tritium Foundry · <a href="https://zenodo.org/record/21746157">10.5281/zenodo.21746157</a>    |
| PAPER II  | Aegis & MetroVolt — The Generator · <a href="https://zenodo.org/record/21746479">10.5281/zenodo.21746479</a> |
| PAPER III | High-Field REBCO Magnets & Tape · <a href="https://zenodo.org/record/21842514">10.5281/zenodo.21842514</a>   |
| PAPER IV  | Direct Energy Conversion · <a href="https://zenodo.org/record/21842864">10.5281/zenodo.21842864</a>          |
| PAPER V   | AI, ML & Quantum Control · <a href="https://zenodo.org/record/21842371">10.5281/zenodo.21842371</a>          |

Explore the whole programme live: the interactive [3-D model](#), the in-browser [physics validator](#), and the [companion film series](#).



KRONOS FUSION ENERGY

# *The HVDC-Native Power Interface*

Explore the science — live, in your browser.

[kronosfusionenergy.com/3D-Model](https://kronosfusionenergy.com/3D-Model)

[kronosfusionenergy.com/physics-Validation-Simulation](https://kronosfusionenergy.com/physics-Validation-Simulation)

[Companion film series · YouTube](#)

[Zenodo · DOI 10.5281/zenodo.22132211](https://zenodo.org/doi/10.5281/zenodo.22132211)